

CHAPTER EIGHT

Specific Strength- Training Methods

By *specific* we mean methods that mimic in some ways the joint angles, forces, muscle-recruitment patterns, and muscle-contraction speeds as well as the metabolic demands of your sport. These specific methods are meant to convert whatever level of general strength you have into sport-specific strength.

In this chapter we will discuss methods of developing both more power and more muscular endurance. These methods can be used by anyone and they will have positive performance benefits. However, if you have deficiencies as discussed in the previous chapter on basic strength, we highly recommend you correct those deficiencies before using the following methods.

The biggest limitations to performance in these mountain sports comes from the need to overcome the force of gravity in climbing up hills. On the uphill is where we see the greatest difference in speed between the elite and the amateur. If you can improve the power and/or the muscular endurance of your uphill propulsion, your performance will benefit directly.

Strength

Uphill running and skiing performance can be improved by increasing the specific strength. High levels of this quality will allow higher levels of the all-important muscular endurance. One of the most sport-specific strength exercises for this is what we term Hill Sprints. There is a full description of Hill Sprints (for runners) or Hill Bounding (for skiers) in the Methodology Section on page 179. Adding poles will help simulate skiing uphill. Include Hill Sprints/Bounding starting in the early Base Period of training for any intermediate and higher-level athlete if basic

strength is adequate as defined in the preceding chapter. If you have already developed a high level of uphill leg strength from previous training, you can consider moving straight to the ME workouts (below). Otherwise it will benefit you to include a few weeks of this type of strength and power work. You can also add occasional Hill Sprint workouts throughout your full training plan to maintain this specific strength. However, if faced with a choice due to time constraints, shorten or omit Hill Sprints to allow more time for a muscular endurance building period.

Muscular Endurance

As we've stated earlier, high levels of strength are not important for the sports covered in this book. However, a particular strength-like quality called muscular endurance is very important. In fact, it is one of the major components in determining endurance performance. As a reminder: This quality is defined as the ability to do many repetitive movements with a *relatively* high percentage of the maximum strength. By *relatively* we mean that the more repetitions you are doing, the lower will be the percentage of your maximum strength. The immediate effect you should feel during these workouts is fatigue localized in the working muscles. While anyone can benefit from this method, the best results will be seen by those athletes with high strength and aerobic capacity. It is tempting to overuse muscular endurance training because the performance benefits are felt quickly and dramatically. When using muscular endurance workouts, it is critical that you do a high volume of basic aerobic training in Zone 1 for maintenance.

There are several ways of improving muscular endurance. We will show you workouts we have used with many types of athletes with universally positive results. As you read through these, notice that despite their very different methods they share the one component that makes them cause local muscular fatigue: extra resistance. By artificially imposing greater muscular loads during training than will be encountered during the event, we can force local muscular endurance of the working muscles, as opposed to cardiovascular fitness, to be the limitation. In our examples, we do this by either the use of added weight or by using a very steep hill.

DO YOU NEED MORE MUSCULAR ENDURANCE?

As mentioned earlier when discussing high-intensity training, Zone 3 and Zone 4 workouts have a powerful muscular endurance training stimulus. Recall that Kílian uses his long, steep uphill Zone 3 runs as muscular endurance training. The effectiveness of any high-intensity session is highly dependent on the athlete having adequate levels of muscular endurance. Kílian has very high levels of local muscular endurance. This allows him to sustain these high intensities for the full duration of the workout. However, many endurance athletes do not have



Oskar Irsara and Raffaella Rungger on perfect singletrack in the Italian Dolomites.
Photo: Dan Patitucci

sufficiently high levels of local muscular endurance. Do you feel yourself slowing in workouts, especially those under two hours when you should not be running out of glycogen stores? Do you feel the heavy, dead legs that lose their bounce in long workouts? Do you feel that low-level burning and acute fatigue in your legs during high-intensity training? Do the downhills beat your legs up? If so, you need to improve your muscular endurance.

The Frontier Fiber Model

The slowing you feel with that leg fatigue is caused by the muscle fibers that are allowing the high speed in the first place dropping out of the pool of recruited fibers. Remember, these are faster twitch fibers than the ones used at slower speeds. Every increase in speed requires your brain to recruit more of these more powerful fibers. These can produce greater force, so are needed to sustain higher speeds. The problem is that by virtue of them having higher power, they have less endurance. They are right at the frontier of your endurance at that speed. Every speed will have its *frontier fibers*: that last group (the highest power bunch) of fibers to be recruited by the brain for this speed. The frontier fibers needed to sustain a pace you can hold for three hours have very good endurance and will



not become fatigued until the run exceeds three hours. Whereas the group of frontier fibers needed to sustain the pace during a VK race or Zone 4 interval session have much more power, but much worse endurance so can fatigue in a few minutes. As you go up the fiber power scale, your endurance drops to the point that the fastest FT fibers (used to sprint) will fatigue in seconds.

These frontier fibers' endurance is, to a very large extent, what limits you. One of the main reasons traditional endurance training methods have included Zones 3–4 workouts is to recruit these frontier fibers and cause them to develop more endurance. If they fatigue too quickly and drop out of the pool of recruited fibers, they are no longer getting the desired training effect. You can tell this is happening when not only do you slow but your heart rate drops despite the increased effort. Do you feel that your legs give out before your breathing during high-intensity workouts or races? If so, you are limited by the local endurance of the working muscles and should consider adding this type of training to your program.

Along with traditional Zone 3 and Zone 4 training, we'll describe workouts you can use to specifically target local muscular endurance. You risk too much of a good thing if you use these workouts too close together or too close to conventional Zone 3 and Zone 4 interval training. Chances are good you will feel these workouts in your legs for a few days. For that reason, if you are new to this form of training it's best to use these workouts in the Base Period as preparation before starting Zone 3 and Zone 4 aerobic interval work. They can also be alternated with Zone 3 and Zone 4 interval workouts. But be sure to keep these separated by 72 hours and maintain a high volume of Zone 1 aerobic base training during this time.

As with all training, especially higher intensity, there is no single rule that works for everyone regarding how much or how often to do these. But, for someone just getting started with this type of training, one day per week will be plenty. Those with more experience with high-intensity training or higher muscular endurance can often manage two workouts in a week before it will begin to negatively affect the volume of aerobic training. We recommend starting these workouts with little to no added resistance so that you can learn the movements and develop the strength and elasticity in the muscles and tendons to avoid injury. After that you will want to be very gradual in ramping up any loads.

While there are many ways to improve muscular endurance, many athletes have had many years of great exereinces with the methods we describe. These workouts have an almost-immediate effect on uphill endurance. If you do not feel improvement to your hill climbing from week to week while doing these, there is a good chance that you are doing too much rather than not enough. You may not be getting sufficient recovery or easy aerobic training during this time.



Swann Juillaguet races while secured to a fixed rope on day four of the 2017 edition of the Pierre Menta skimo race. Photo: Jocelyn Chavy

GYM-BASED MUSCULAR ENDURANCE (ME) WORKOUT

Not everyone who loves mountain sports has easy access to mountain terrain. Or maybe you're a runner who wants to increase muscular endurance during a dark, cold, and snowy winter. Author Johnston first experimented with gym-based ME programs in 1991 when working with an elite cross-country skier leading into an Olympic year. Since then we have developed countless others for different athletes and different sports. We have learned a lot over the years through this experience. One of the most important lessons: Don't let the non-sport-specific nature of these workouts fool you. They have proven very effective at improving uphill and downhill running and skiing performance, especially for those without ready access to mountains.

GYM-BASED ME WORKOUT

Warm-up

A good warm-up is essential for this workout. You need to be sweating before you start the workout. Here is a warm-up we find effective:

Do 200 total reps of your choice four or five different core exercises without stopping. Pick some that you like or find challenging. Try to do a minimum of 20 reps of each exercise before moving, with no rest, to the next exercise. Then rev up your aerobic system. Fifteen minutes should do it. How you do this is not so important: bike, run, stairs, rowing, or a mix. Importantly, finish with two to three minutes of moderately hard effort (Z3).

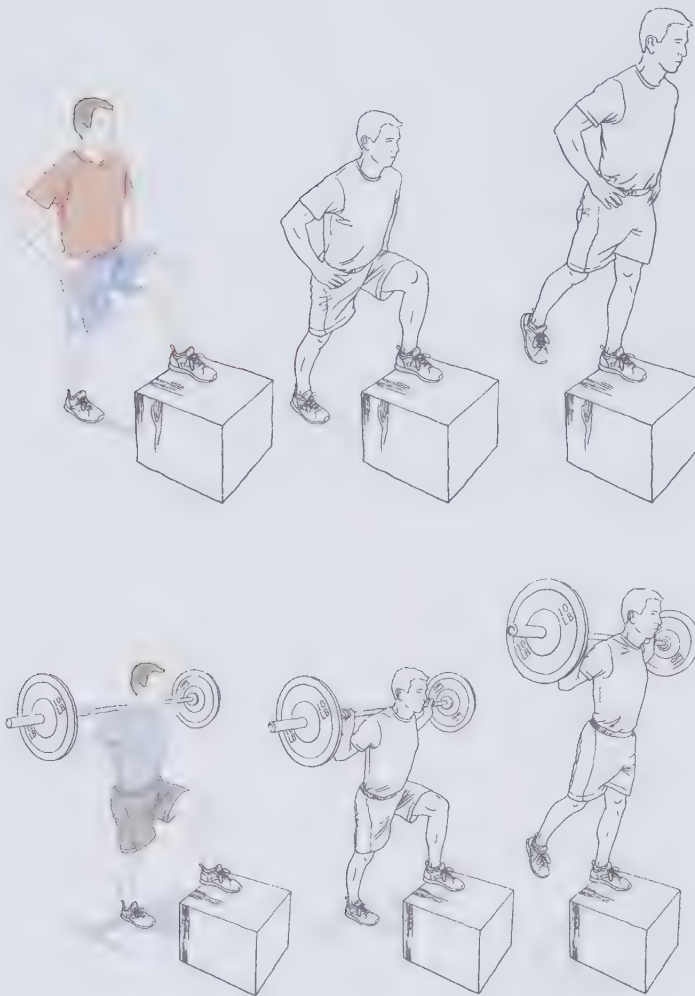


Fig. 8.1 Box Step-Up

Stand about one half of your stride length in front of a box or step that is just below knee height. Step up onto the box mainly using the front leg to pull you up with minimal assist from the rear leg. Maintain proper knee tracking over your toes throughout this exercise.

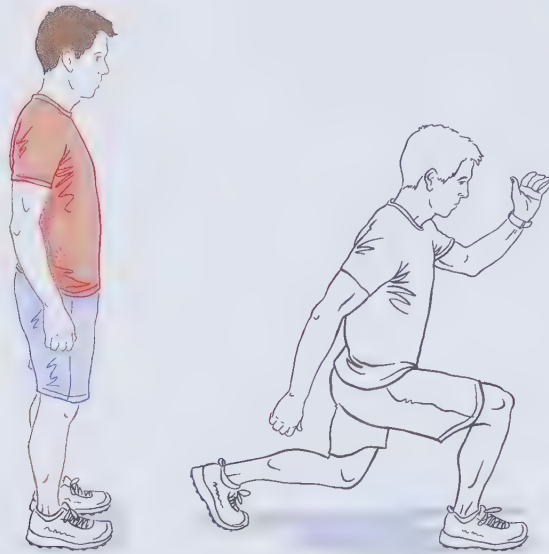


Fig. 8.2 Front Lunge

From a standing position step forward about half your body length dynamically so that the front leg catches most of your weight. Keep the knee behind the toes of that front foot.

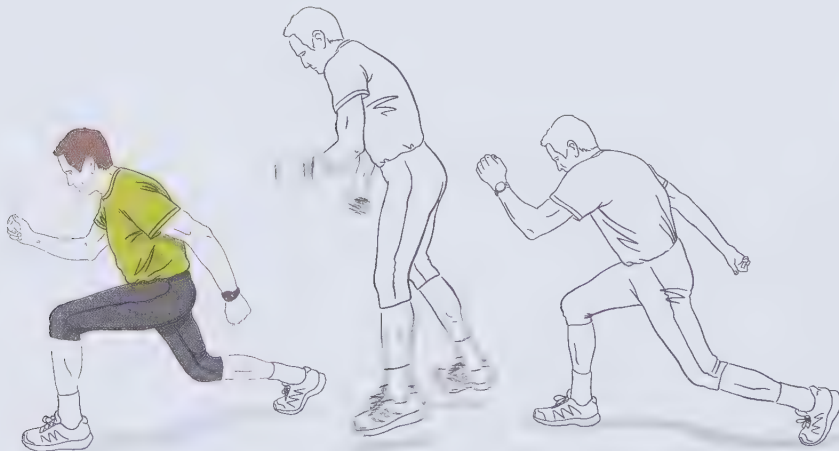


Fig. 8.3 Split Jump Squat

From a front lunge stance spring upward and switch feet before landing with the opposite foot forward. You can add resistance best with the use of a weight vest or weighted pack. Count reps on just one leg. Do, ten reps on each leg.

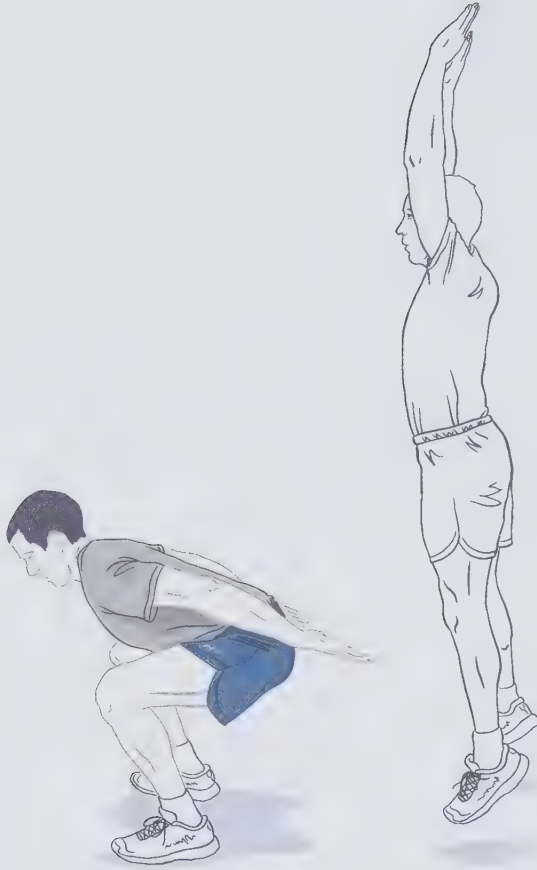


Fig. 8.4 Squat Jump

From a squat position spring upward as explosively as you can. Swinging your arms powerfully will increase the load on your legs and help make this more dynamic. Land quietly, absorbing the load using your toes and knees.



Kilian Jornet doing steep (and technical) sprints above the Val Veny, Courmayeur, Italy. Photo: Sebastián Montaz-Rosset

Workout

- 6x10 reps each leg of box step. Rest one minute between sets. Ten reps with one leg before switching.
- 6x10 reps each leg front lunge. Rest one minute between sets. Ten reps with one leg before switching.
- 6x10 reps each leg split jump squat. Rest one minute between sets. Alternate front legs each jump.
- 6x10 reps squat jump. Rest one minute between sets.

Complete all six sets of each exercise, taking one minute of rest between sets. Take no more than two minutes' rest between exercises. The tempo of the exercise should be about one repetition per second.

Workout Progression

- First and second workout use body only weight.
- Third and fourth workout add 10 percent of body weight.

- Fifth workout do eight sets of each exercise add 10 percent of body weight.
- Sixth and seventh workout do six sets add 10 percent of body weight and reduce rest to 30 seconds.
- Eighth workout do eight sets with 10 percent body weight and 30 seconds' rest.
- Continued longer-term progression:
- Ninth workout do six sets with 15 percent of body weight resting one minute.
- Tenth workout do six sets with 15 percent of body weight resting 45 seconds.
- Eleventh workouts do six sets with 20 percent of body weight resting one minute.
- Twelfth workout do six sets with 20 percent of body weight resting 45 seconds.
- Use a weight vest or a barbell on shoulders or hold dumbbells for added resistance.

The percentage of body weight listed above are suggestions. While these seem to work well for most athletes, you may need to use more or less depending on your basic strength. Start lighter and increase the weight only as you feel the effects of the previous workout. Make adjustments gradually and err on the conservative side. These workouts can be spaced out by as much as ten days but if you go more than twelve days, repeat the previous workout rather than moving ahead to the next one. You do not need to go through the full eight workout progressions to get the benefits. Or you may find that you have time to add more workouts to see bigger gains during your off-season.

OUTDOOR ME WORKOUT

This workout resembles a conventional Zone 3/Zone 4 uphill interval workout. It works well for those with some big and steep hills nearby and without easy access to a gym or weights. We've used it with very good results for both skiers and runners. There are several ways to accomplish this training and a fair bit of latitude exists, so don't get too hung up on the details. Keep the big picture in mind: Make your legs be the limitation of your ability to sustain the work intensity. Make your legs tired! Preferably before your breathing shuts you down.

The biggest challenge for most people is finding a steep enough hill. Look for a hill that is in the range of 20 to 50 percent in gradient. The steeper the hill, the easier it is to feel that your legs are the limitation in your rate of climb, not your breathing or heart rate. Several methods exist to increase the muscular load on your legs: steepness, stride length, and added weight. Any and all can be used

depending on what kind of terrain you have available and how strong your legs are to begin with. Let's discuss how you might incorporate these three factors:

- **Steepness:** This could mean running or hiking straight up the fall line at a gradient on excess of 20–50 percent. Most people will call a trail steep when it exceeds 20 percent but for these workouts to be most effective, steeper is better. This is what Kílian is referring to when he talks about doing long, steep uphill in Zone 3 as a muscular endurance workout. Settle into a pace you can sustain for many minutes with a subtle, low-level “burn” in your legs.
- **Stride length/bounding:** Use a long, bounding stride. This adds a significantly high muscle load and cannot be sustained for long. This is what Kílian is referring to when he mentions doing two minutes of long strides followed by a short rest until his legs recover, then two minutes of short strides. If you are bounding aggressively, two minutes is a long time. Be advised that you may want to start with ten to twenty seconds of bounding or until your legs are burning, followed by a recovery to get your legs back, then the short-stride/high-tempo running until your breathing and heart rate hamper you. Then, repeat many times.
- **Additional weight:** If you don't have access to steep terrain, you can simulate the same by adding weight to a small pack and doing either of the above methods. Start by carrying no more than 10–15 percent of your body weight. Add more only after you have seen the effects of this amount.

One session of this workout each week to ten days is adequate during the Base Period; otherwise it can impact the overall volume of training negatively. It is important to maintain a high volume of Zones 1–2 training during this phase.

OUTDOOR ME WORKOUT

Warm-up

Running 15 minutes with gradually increasing intensity so that the last two to three minutes are in Zones 3–4.

Workout Progression

As with the gym-based version of ME workout, be prepared to get sore and have tired legs for a couple of days. You should feel the hills on your normal aerobic workouts getting easier within only two to three workouts of this type.

Using the traditional Z3 uphill method:

Start this workout progression by running or hiking (which you choose depends

on the steepness and your muscular endurance) up the hill for eight to ten minutes with that low-level leg burn. Repeat two times, taking a five-minute moving recovery between reps. If you need to walk back down between reps because you do not have a long enough hill, that's fine. This can build to a full 40 minutes of steady uphill grinding.

Using a weighted pack for the above method:

This can be particularly effective for ski mountaineers who will normally be carrying a heavy pack and moving more slowly upward. It can also be used by those who do not have access to steep hills. Don't discount the usefulness of the fire stairs in tall buildings for this type of workout.

If you use the bounding variation:

Start with 10x10 sec of bounding up a steep trail. Between the bounding parts, walk back down until your legs are recovered and then repeat. There is no one prescription. For the aerobically fit who recover fast, this workout might look like 15x15 sec bounding with about one minute of running or walking down recovery (depending on steepness) between repetitions. This workout could progress to 20x20 sec of bounding with two minutes of recovery running. For the less fit, start with something in the range of 10x10 with two minutes of active recovery.

As with the gym-based muscular endurance training, if you do not see performance improvements from week to week, you are probably doing too much and not allowing enough recovery.

Upper-Body Training for Skimo

A good way for more advanced skiers to boost their performance is to include special upper-body muscular endurance training into the Base Period. Because this training only minimally involves the legs, it's easy to add arms-only workouts even into a heavy off-season of running training. You may find that they even aid your legs' recovery. We present two straightforward methods of doing this training below, but variations on these are easy to imagine. With these workouts you will be able to feel very strongly the muscular endurance limitation that we hope you feel doing the leg muscular endurance work. The muscle mass you use for this single-poling is small enough that it will not tax your heart's ability to supply oxygen. The limit you feel will be quite obviously in your arms and shoulders.

(C) ROLLER SKIING

Roller ski training will likely only appeal to a minority of readers who seek the best training methods; we include it here for the sake of completeness. Roller skis provide a good simulation of snow skiing. When used with good technique they will improve your weight transfer, enhance your glide on each stride, and train

your muscles in a ski-specific way. One drawback is that most paved roads are not as steep as the terrain you will be skinning up. However, using your arms only, even on modest uphill, will have the desired upper-body muscular endurance training effect. You will need classic roller skis and carbide-tipped ski poles. Be forewarned: Roller skis require a significant learning period. They have no brakes, so downhills can be very dangerous.

Early Base Period roller ski upper-body ME workouts can be done on relatively flat terrain by simply using the same poling motion that you normally use when skinning. Adopt a slight crunch position (a C shape) in your upper body to give you more muscle involvement for the poling motion.

Beginner

Those with little Nordic skiing background may find even poling on the flats for more than a couple of minutes very fatiguing. If so, this should be a clue that you could see big gains in uphill speed with even a small amount of time devoted to this type of training.



Fig. 8.5 Roller Ski Arms Only

This exercise is a great way to specifically train the upper-body muscular endurance needed for skimo. Hold the upper body and feet still and work with just the arms. You can adjust the load by increasing the gradient of the hill.

Example Workout

Start with continuous single poling for ten to fifteen minutes on the flats to get the feeling of this type of workout. After a few workouts, you may feel ready to increase the intensity. A good way to do that would be to do 5x1 min at a moderately hard effort on a very gradual uphill with a two-minute recovery. Progress to 3x10 min with five-minute recovery periods where you are still skiing but very easily again on the flats or a very gradual uphill. Select the grade to suit your upper-body strength.

Advanced

Those with more upper-body strength and endurance should do these same single-poling ME workouts but on steeper grades. Mix steep and short repetitions with longer, lower-angle reps.



Author Scott Johnston uses a Concept 2 Ski Ergometer for a ski-specific upper-body muscular endurance workout in the Uphill Athlete World Headquarters gym. Photo: Seth Keena

These types of workouts can be included twice per week in the early Base Period as a second workout on days with leg-focused workouts. They can be continued on to snow and as a maintenance session or during breaks in the racing calendar.

SKI ERGOMETER

A common training tool for cross-country skiers is an ergometer, like a rowing machine, that uses ropes and resistance to simulate the poling action.

What follows is an example of a stand-alone upper-body ME workout progression that can be used as a second workout on days that involve a leg-oriented workout.

Progression:

- Week 1: 5 min easy single-poling finished off with 1 min at a moderate effort.
- Week 2: 5x2 min hard effort with 2 min active recovery like easy running/hiking on a treadmill.
- Week 3: 6x2 min with 2 min active recovery.
- Week 4: 7x2 min with 1:45 min active recovery.
- Week 5: 7x3 min with 2 min active recovery.
- Week 6: 8x3 min with 1:45 min active recovery.
- Week 7: 8x3 min with 1:30 min active recovery.
- Week 8: 8x3 min with 1 min active recovery

Summary

Use the Base Period for developing strength and muscular endurance according to your needs as an athlete. If you are deficient in strength according to your assessments, then it may be that the first cycle of training you do using this book will include only general strength training. As you train more and become stronger, it will benefit you to add Specific Strength and ME training during the Base Period.